IJC445 Coursework 2025-2026

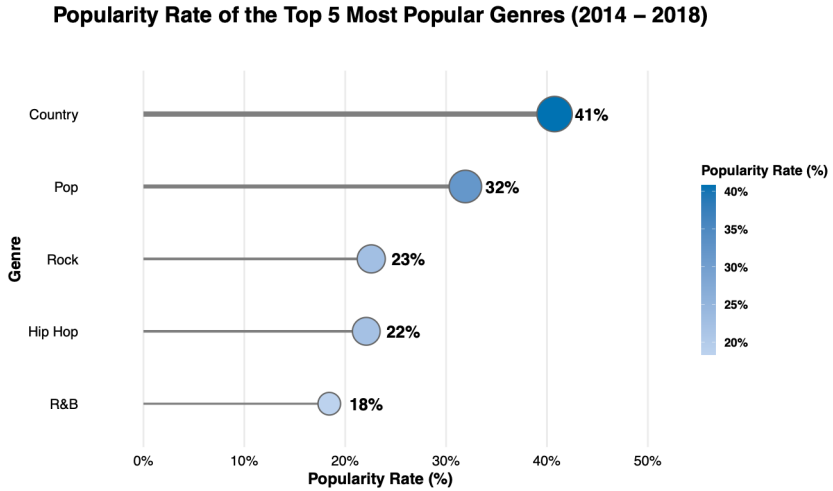
Exploring the Relationship Between Metadata Features and Song Popularity in the Streaming Era (2014 – 2018)

Programme: MSc Data Science, University of Sheffield

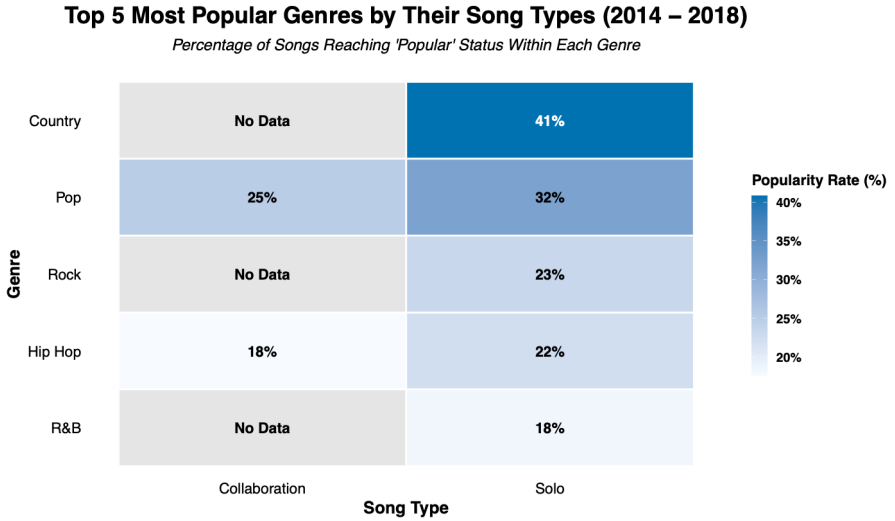
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Composite Visualisation

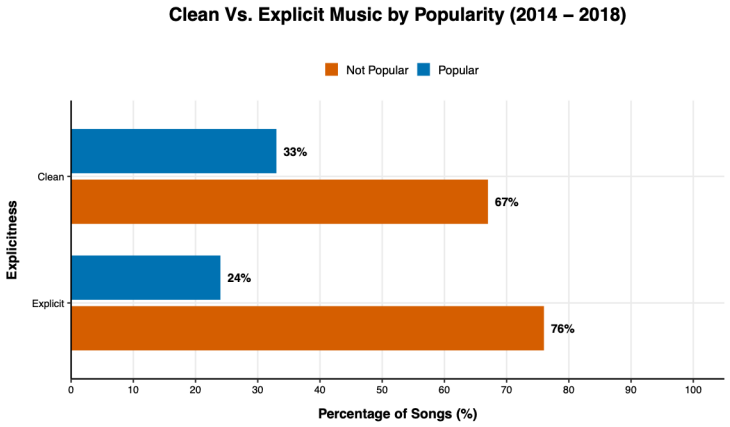
(Figure 1)



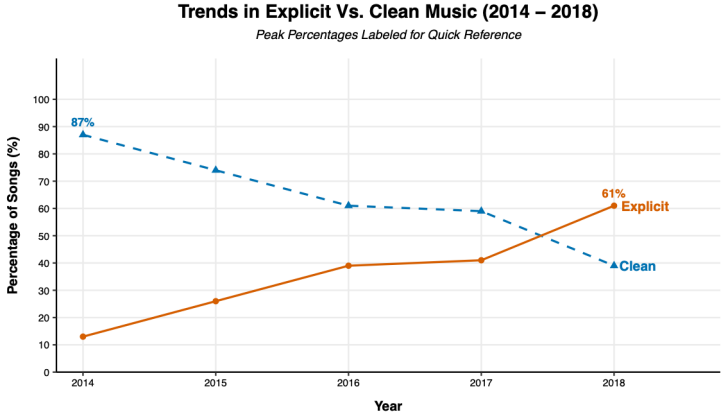
(Figure 2)



(Figure 3)

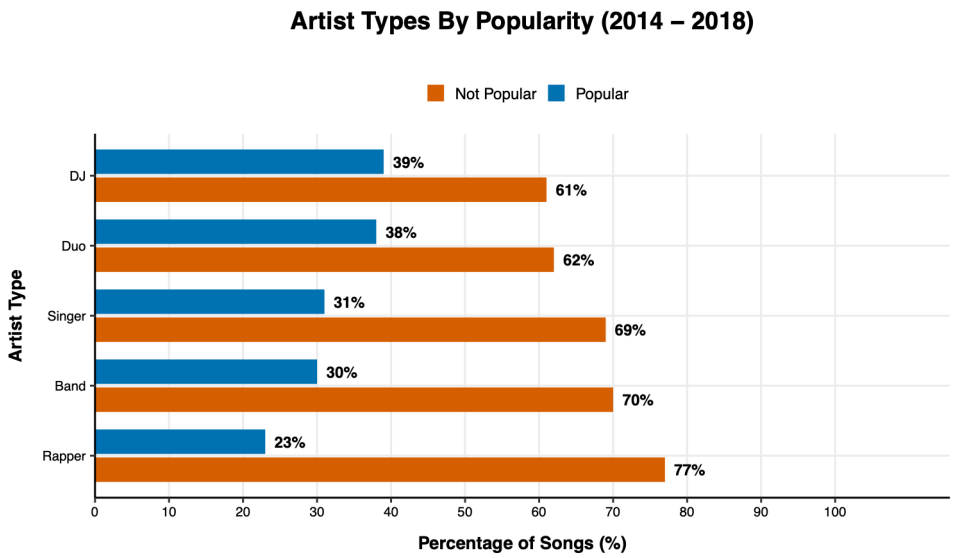


(Figure 4)



Composite Visualisation

(Figure 5)



1. Knowledge Building

1.1 Topic and Question

In 2023, music labels invested a record $8.1 billion USD in artist acquisition and development, yet approximately 80-90% of artists did not achieve commercial success (Rechardt, 2025). Between 2014 and 2020, only 1% of artists accounted for nearly four-fifths of all streams in the United Kingdom (Hesmondhalgh et al., 2021). These figures highlight considerable demand among industry stakeholders to identify the factors driving musical popularity. While most machine learning research focuses on acoustic attributes (Lee & Tu, 2025; Zhao et al., 2023), a substantial research gap persists regarding the explanatory power of metadata.

This study employs data visualisation techniques to examine the relationship between metadata features (genre, song explicitness, artist type, and song type) and song popularity in the streaming era. This study uses the MusicOSet dataset (Silva et al., 2019b) and focuses on songs from the Billboard Hot 100 charts between the years 2014 and 2018. The binary popularity metric *is_pop* is utilised in this study to address the absence of non-chart-topping tracks within the dataset.

1.2 Composite Visualisation

Figures 1 and 2 present the top 5 genres with the highest counts of popular songs. Figure 1 shows that country music leads, with 41% of its songs classified as popular. However, the lollipop chart suggests that genre is a weak indicator of song popularity. Despite notable differences in popularity rates between R&B and country (a 23% difference),

Hip-Hop and country (a 19% difference), and rock and country (a 18% difference), the majority of songs within these genres are not popular, with non-popular rates ranging from 59% to 82%. Figure 2 demonstrates that solo song types are more prevalent in popular tracks than collaborations in these leading genres. Although genre appears to exhibit a limited relationship with song popularity, the data indicates that music labels may benefit from prioritising country solo artists, as they display a higher success rate. Since the Billboard Hot 100 charts reflect American listening habits, these findings may be attributed to the substantial listener base for country music in the United States, where up to half the population regularly engages with the genre (Bates et al., 2020).

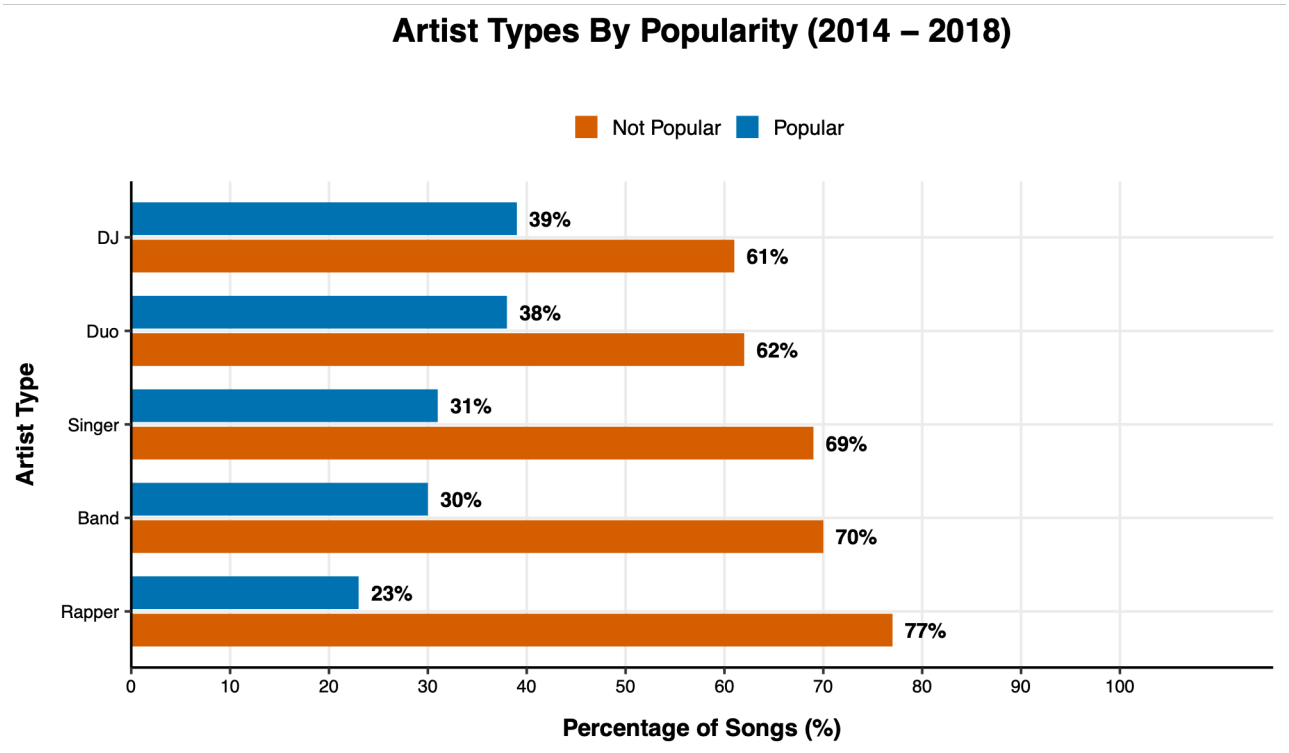
Figure 3 shows that while song explicitness may relate to popularity, its effect appears to be relatively limited. The slight variance (9%) between the popularity rates of clean (33%) and explicit songs (24%), along with the high proportion of non-popular songs in both categories (67% and 76%), suggests that song explicitness is not a primary indicator of popularity. Figure 4 further reveals that, despite a substantial industry shift toward explicit music (61% of charting songs in 2018), this trend has not led to proportionately greater popularity of explicit tracks. These findings reinforce the potentially weak relationship between song explicitness and popularity. As shown in Figure 3, the higher popularity rate of clean music (33%) compared to explicit music (24%) may be attributed to psychological effects on listeners. Research suggests that lyrical content can influence emotional regulation: violent lyrics may increase hostility, while prosocial lyrics can reduce aggressive outcomes (Anderson et al., 2003; Greitemeyer, 2011).

Figure 5 indicates that artist type exhibits a limited relationship with song popularity. Although the difference in popularity rates between DJs and rappers is notable (a 16%

difference), the proportion of non-popular songs within artist types remains high (61% to 77%). DJs display the highest percentage of popular songs (39%) among the artist types studied. One study suggests that DJs' frequent incorporation of other artists' work may enhance their visibility across multiple fanbases, potentially contributing to higher popularity rates (Wi & Lee, 2021).

In summary, the visualisations indicate that metadata features are likely weak predictors of song popularity, although further statistical analysis is required to confirm these relationships. This may explain the academic preference for audio-feature analysis. Based on the observed trends, the features ranked from most to least notable are: (1) genre, (2) artist type, (3) song explicitness, and (4) song type. To conclude, when considering metadata features, music labels seeking to optimise investment may benefit from prioritising country or DJ soloists with clean lyrics.

(Figure 5)



2. Theoretical Frameworks

The following section focuses on Figure 5:

1. **Ask a Question:** To what extent is artist type associated with song popularity in the streaming era?

2. **Search for Information:** The data source used to create visualisations must be open-sourced and easily accessible (Kirk, 2012). The MusicOSet dataset (Silva et al., 2019b) is publicly available on GitHub and offers unrestricted access, with two file types (CSV and SQL) for ease of use. Additionally, the data source should be reliable. The MusicOSet gathers data from industry leaders such as Billboard, Spotify API, and Genius.

3. **Structure the Data:** Cleaning and analysis were conducted using RStudio version 2025.09.0+387. First, filtering was performed, selecting data only from 2014 – 2018, as these are the most recent streaming era years within the MusicOSet. Then, 64 rows with missing artist type values were removed. According to Ferster (2013), one of the most crucial components of structuring the data is identifying the quantitative and qualitative variables required for the question and putting them into sub-categories. Artist type is the nominal categorical independent variable, and song popularity (*is_pop*) is the binary categorical dependent variable.

4. **Envision the Answer:** An exploratory data analysis was performed using RStudio functions, and a table was created showing the percentage and count of popular

and non-popular music by artist type. A grouped bar chart was chosen because the question requires a side-by-side comparison of subcategory frequencies within a main set of categories.

5. **Represent the Visualisation:**

Geometry: The bar is selected due to the categorical nature of artist type. Bars are typically used to show the frequency of a categorical variable, whereas points or lines are generally used for continuous variables (Pallant, 2020). Also, bars, including grouped bar charts, induce the least cognitive load compared to other graphs for categorical variables, such as donut or pie graphs (Hill, 2025). Grouped bars were chosen instead of segmented bars. Cognitive interpretation is improved when viewing a grouped bar chart rather than a segmented bar chart (Talbot et al., 2014).

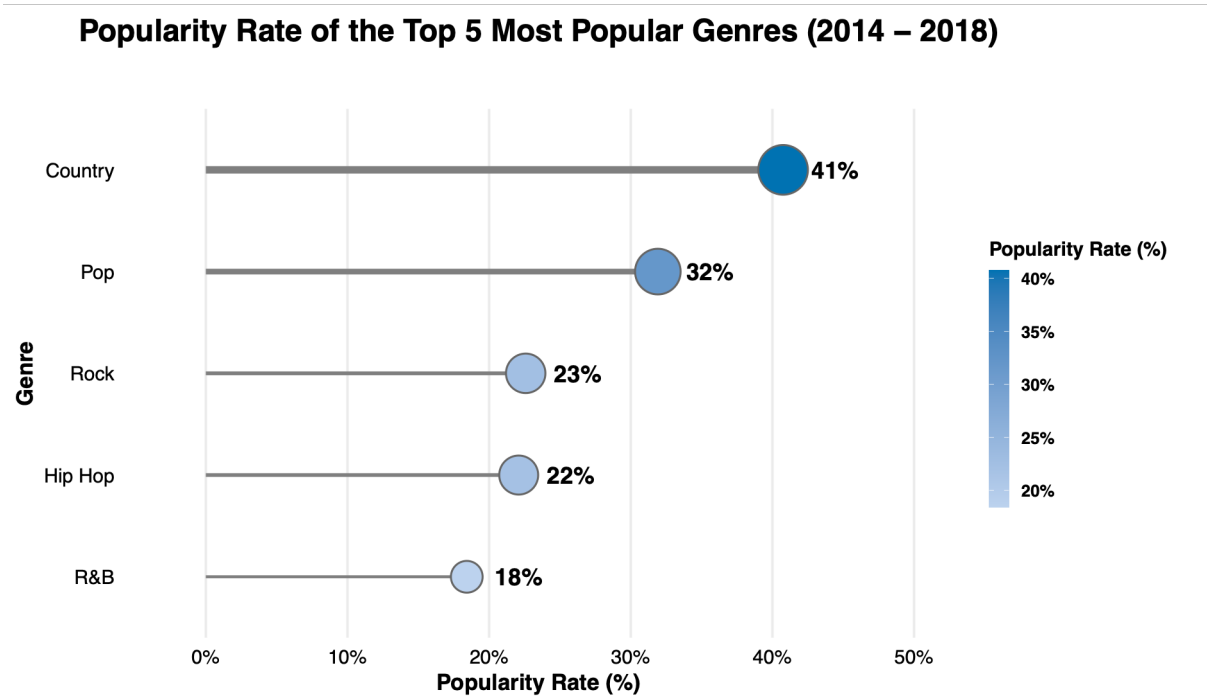
Aesthetics: For instance, the bars are coloured by popularity status to distinguish between subcategories effectively. The Okabe-Ito pallet was utilised. This pallet is well documented as being accessible for colourblind people (Zeileis & Murrell, 2023). Additionally, the artist types are listed on the y-axis rather than the x-axis, where they are usually located. This aids in readability.

Coordinate System: The grouped bar chart utilises the Cartesian coordinate system. A grouped bar chart was chosen over a pie chart and a polar coordinate system, as perception is significantly higher when assessing length compared to angles (Cleveland & McGill, 1984).

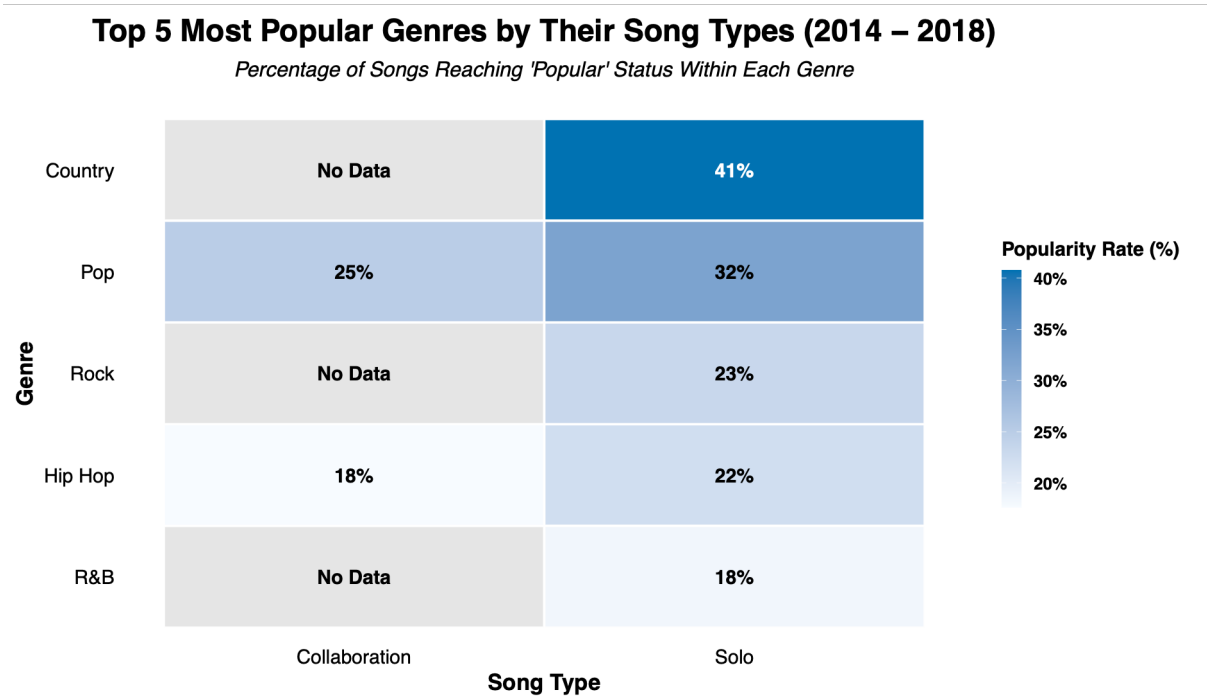
Theme: Percentage labels were added next to each bar, so the viewer does not have to keep referring back to the x-axis to find the exact value. Adding numeric labels to bars reduces cognitive load (Kong, 2012). Furthermore, light grey gridlines were used to help ease visual strain. Sufficient gridlines can assist the eyes in tracking the percentages between categories.

6. **Tell a Story:** Bates (1989) discusses the 'berry picking' metaphor, in which people tend to select only pieces of information from various sources due to selective attention. Therefore, it is imperative that visualisations concisely and clearly showcase only the most essential information. Figure 5 contains only one independent variable along with the dependent variable. Finally, a story must be effectively communicated to its audience. The audience for this study could be music label executives, musicians, academic researchers, or the public. Therefore, most of the audience would not have a background in statistics. By using the elements of the grammar of graphics correctly, the visualisation is presented in a way that minimises cognitive load and promotes sufficient comprehension.

(Figure 1)



(Figure 2)



3. Accessibility

The following section focuses on Figures 1 and 2:

Accessible visualisations are defined as graphs that incorporate design features which are inclusive of people with various impairments, including cognitive (Devane et al., 2024) and visual disabilities (Kim et al., 2021). The following section discusses how Figures 1 and 2 are inclusive of colour-blindness and other visual disabilities, as well as cognitive disabilities (such as dyslexia). This section will also cover how these visualisations highlight the importance of creating diagrams that are cognisant of physical disabilities (such as migraines), as well as economical.

Both graphs utilise the blue colour (#0072B2) in the Okabe-Ito palette. The Okabe-Ito palette is one of the most accessible colour schemes for colourblind people (Zeileis & Murrell, 2023). Additionally, since both graphs use various shades of blue, individuals with achromatopsia, or total colour blindness, can identify the different cells and lollipops, due to the varying brightness levels. The percentages in the heat map cells and lollipops, also enhance readability. Viewers can easily decipher the visualisations regardless of the colours of the cells or lollipops.

Furthermore, the heat map and the lollipop chart contain black text on a white background. This creates high contrast, maximising legibility. Several studies discuss how high contrast improves readability for people with various visual impairments (Cheong et al., 2008; Pizano, 2016). The percentage labels on the cells and next to the lollipops reduce eye tracking. Viewers do not need to keep referring to the x-axis or legend, as the percentages are easily accessible. Therefore, these graphs are accessible to individuals with macular

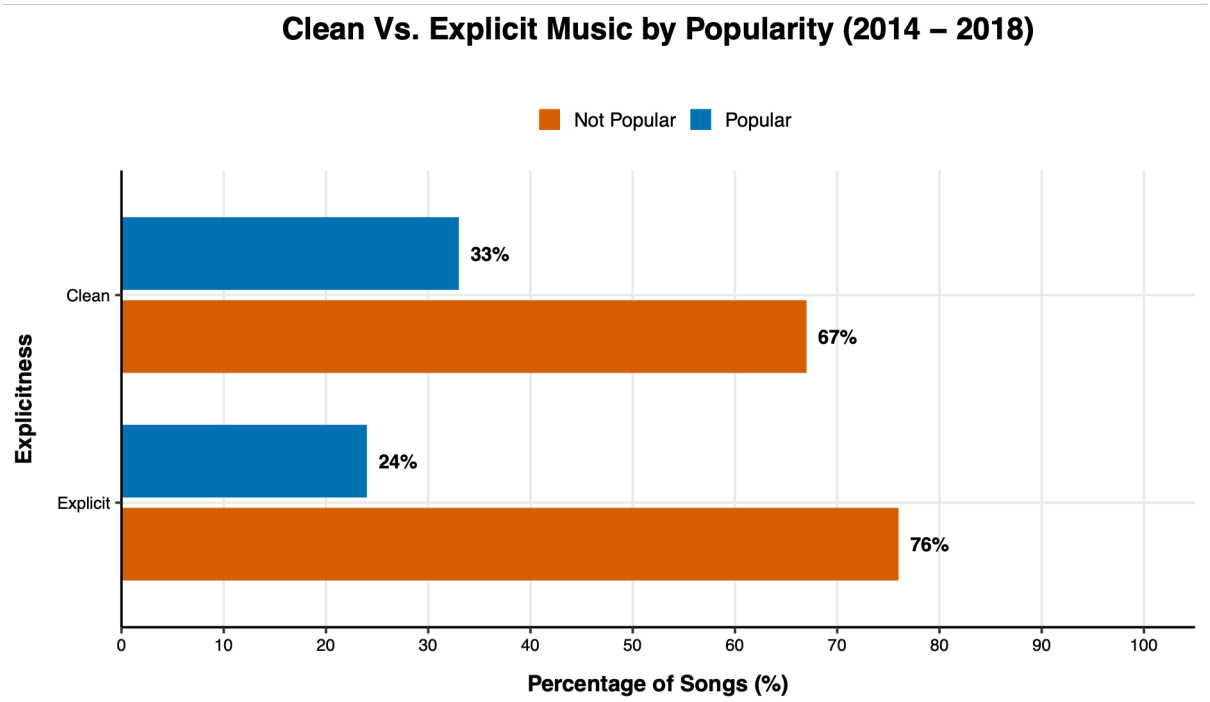
degeneration. Macular degeneration leads to central vision blindness, causing significant reliance on peripheral vision (Burge et al., 2016). Through the proper use and placement of the percentage labels, all the crucial information is condensed into a small area on the graphs. This is ideal for people with a limited field of vision.

Figures 1 and 2 also use a sans-serif font. Sans-serif fonts are the most legible for people with dyslexia (Bilotti et al., 2025). In addition, the heat map cells are differentiated by the white border around each of them. There are gaps between the lollipops as well. People with dyslexia are highly sensitive to crowding, and white space helps to significantly reduce this issue (Marinus et al., 2016; Zorzi et al., 2012).

Since both visualisations use various shades of blue, the differences in brightness allow these graphs to be printed in black and white. These graphs are more cost effective, since monochrome (black) ink is cheaper than polychromatic (color) ink. Figures 1 and 2 can also easily fit on A4 paper, making them printer-friendly. Also, an estimated 12% to 15% of Americans experience migraines, with women being disproportionally affected (Cohen et al., 2024). Furthermore, a study found that screen time significantly increases migraine frequency (Montagni et al., 2016). Therefore, because they are printer-friendly, these graphs also assist people suffering from chronic migraines.

In summary, Figures 1 and 2 effectively communicate findings to individuals with visual and cognitive disabilities. They also stress the importance of considering people with physical ailments when creating visualisations. Therefore, these diagrams are highly accessible.

(Figure 3)



4. Visualisation Choice

The following section discusses Figure 3:

4.1 Alternatives

1. Segmented Bar Chart:

This graph shows the whole picture for each category of song explicitness. Unlike grouped bar charts, segmented bars share the same height (100%), so the viewer does not have to mentally verify that the subcategories add up to 100%. Additionally, too much information on a graph creates visual clutter, which reduces perception and readability (Fu et al., 2021; Herman et al., 2000). Segmented bar charts have fewer bars compared to grouped bar charts.

However, Cleveland & McGill (1984) found that length perception is significantly improved when viewing grouped bar charts rather than segmented bar charts. It is more challenging for people to compare segments that do not share a common starting point, as in the top portion of a segmented bar chart. Therefore, grouped bar charts are easier to analyse, as each bar starts at zero.

2. Pie Charts:

The information presented in Figure 3 could have been displayed by two pie charts. One pie chart would show the total percentages of popular and non-popular explicit songs, and the other would show the total percentages of popular and non-popular clean songs. Silvia & Barona (2009) found that individuals prefer curved shapes to angular shapes. This suggests that people may be more drawn to the circular shape of a pie chart than to the angular bars of a bar chart.

As mentioned in the Theoretical Frameworks section of this paper, accuracy decreases when individuals analyse angles, as opposed to lengths (Cleveland & McGill, 1984). Since grouped bar charts have long bars and pie charts have angular slices, this provides further evidence in support of Figure 3.

3. Cleveland Dot Plot:

The results could have been showcased using four dots. The percentage of songs would be displayed on the x-axis, and the categories for song explicitness would be on the y-

axis. The dots would be coloured by popularity status. Similar to the segmented bar chart, the dot plot prevents visual clutter by using four small dots, instead of four long bars.

Studies have found that people associate larger elements of an image or graph with higher values and they are therefore interpreted by the brain as being more important (Bylinskii et al., 2017; Fygenson & Padilla, 2025). Therefore, people may overlook this visualisation due to its small dots.

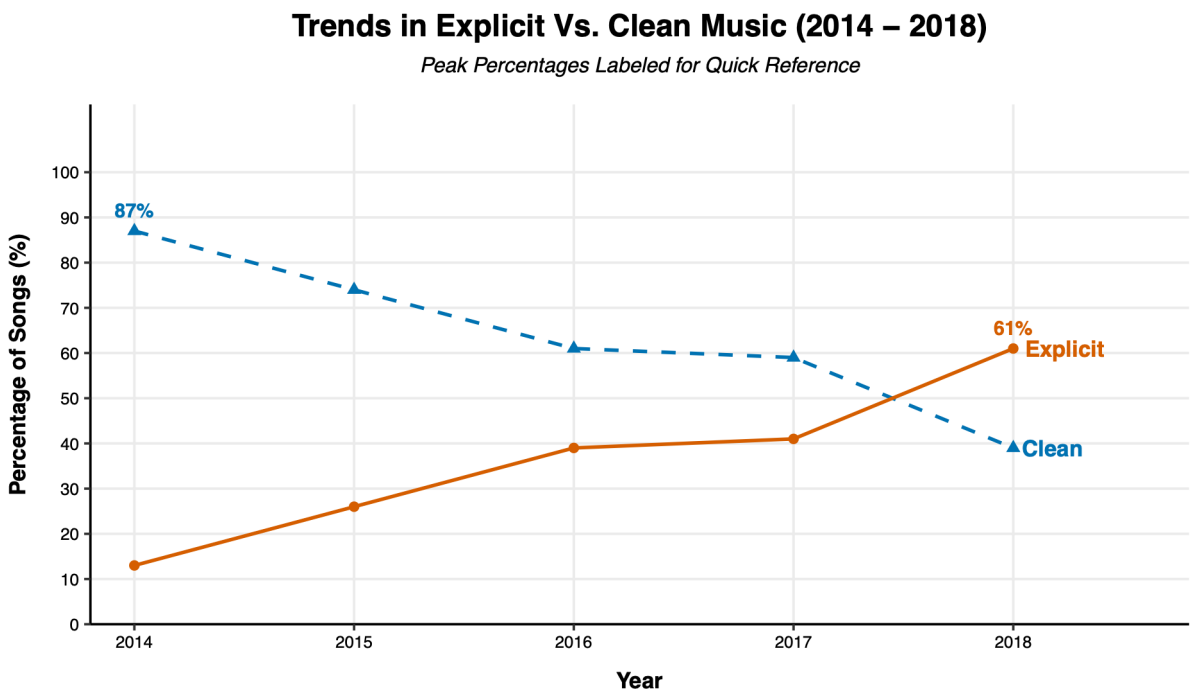
4.2 Justification of the Type of Visualisation

Although the grouped bar chart does have disadvantages, these shortcomings can be mitigated. Having four bars instead of two in a segmented bar chart can increase cognitive load. There is also the dilemma of requiring individuals to perform mental math to confirm that the totals in each song explicitness category add up to 100%. However, length perception is a component of pre-attentive processing (Fusco et al., 2025; Ware, 2013). Therefore, because long bars are used, the visualisation still facilitates gist extraction with minimal cognitive effort.

The grouped bar chart does possess highly useful features that set it apart from other visualisation choices. As discussed previously, having all bars start at zero provides a more straightforward interpretation. Additionally, people are better at processing lengths of bars than angels (Cleveland & McGill, 1984). Even though these issues with pie charts and segmented bar charts could be solved by using percentage labels, the reading and cognitive processing required for these labels would result in higher cognitive load compared to the pre-attentive processing that grouped bar charts allow. Finally, it was also mentioned that

long bars are more attention-grabbing than small dots. In summary, the advantages of the grouped bar chart, combined with its weaknesses and their mitigation, demonstrates that Figure 3 is the superior graph choice for this data.

(Figure 4)



5. Implications and Improvements

Section 5.1 focuses on Figure 4 and Section 5.2 focuses on the composite visualisation:

5.1 Ethical Implication

When viewed in isolation, Figure 4 misleads the public and can lead viewers to make inaccurate conclusions. This visualisation suffers from selection bias. Selection bias occurs when the information gained from the sample size is not reflective of the true population (Lu

et al., 2022). Figure 4 is not representative of the entire dataset because it omits the crucial *is_pop* variable. From the visualisation, viewers are led to the false conclusion that explicit songs are more popular than clean song, when the broader data suggests otherwise.

Kennedy et al. (2016) discusses several elements required for people to interpret a visualisation as being impartial, substantiated, and straightforward. One element is the use of a two-dimensional plane. This instils a sense of integrity, as people can see the whole picture without requiring multiple viewpoints, unlike three-dimensional visualisations. Another characteristic is the use of simple geometric shapes, which establishes structure and resonates with the human desire for order. Furthermore, utilising lines and simple shapes reduces cognitive load and creates a sense of clarity, reinforcing the perceived validity of the information. Another additional element mentioned is having a minimalist layout. This brings clarity, suggesting to the mind that the information in the visualisation is objective. Figure 4 is on a two-dimensional plane. Repetition of shapes is evident in the rectangles created by the gridlines and in the rectangular layout of the graph. Also, the layout is clean, with minimal text. Therefore, viewers can easily perceive Figure 4 as a transparent and objective visualisation.

However, when Figure 4 is examined alongside Figure 3, the viewer is led to the correct conclusion that explicitness is potentially a weak predictor of song popularity. Figure 3 also suggests that clean music is more popular. The combination of the two graphs creates a more well-rounded visualisation.

5.2 Future Improvements

There are multiple improvements to be made regarding the dataset and the composite visualisation. Also, there are avenues for further research. Silva et al. (2019a) mention the issue that most of the songs on the MusicOSet are from the United States. Also, the dataset only contains songs on the Billboard charts, and the most recent music is from 2018. Therefore, the dataset should be adjusted, and future research should include more recent and culturally diverse music. For more accurate results, greater emphasis should also be placed on including non-chart-topping music.

Regarding the composite visualisation, adding another graph showcasing the total percentages of popular and non-popular solo and collaborative songs in the dataset would be highly beneficial. The information gained from the additional visualisation would help illustrate whether song type, as a whole, is associated with song popularity.

Finally, as previously stated, the majority of research has been conducted on acoustic features (Lee & Tu, 2025; Zhao et al., 2023). A substantial research gap remains regarding metadata features. As this study relied solely on descriptive data visualisations and did not apply predictive modelling, it is not possible to establish a definitive link between metadata features and song popularity. Further empirical research is required. In addition, Choudhary et al. (2025) refer to the impact of lyrics on song popularity as being ‘under-explored’.

In conclusion, the composite visualisation strictly adheres to the principles of the grammar of graphics and is highly accessible. Although further improvements could enhance

its depth, the composite visualisation successfully showcases pivotal preliminary insights into how metadata features relate to song popularity in the streaming era.

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Appendix: R Code

The full R code is reproduced below.

```
# Load the libraries. Install these if you don't have them before ru
nning the code.
library(tidyverse)
library(stringr)
library(ggfittext)
library(scales)
library(tidyr)

# ************************************************************
# PART 1. CREATING DATASET AND CLEANING
# ************************************************************

# First, load in the cleaned songs and artists datasets.
# I have named these datasets as "cleaned" because I have altered th
ese
# by deleting columns and/or editing columns.

songs_cleaned <- read_csv("songs_cleaned.csv", col_names = TRUE)
songs_cleaned

artists_cleaned <- read_csv("artists_cleaned.csv", col_names = TRUE)
artists_cleaned

# Everything looks great with these 2 data frames.
# Now left join them together.

songs_artists_combined <- songs_cleaned %>%
  left_join(artists_cleaned, by = "artist_id")
songs_artists_combined

# Now, time to load in the cleaned acoustic features and left
# join it to songs_artists_combined.

acoustic_features_cleaned <- read_csv("acoustic_features_cleaned.csv
", col_names = TRUE)
acoustic_features_cleaned

songs_artists_acoustic_combined <- songs_artists_combined %>%
  left_join(acoustic_features_cleaned, by = "song_id")
songs_artists_acoustic_combined
```

```r
# Now, time to load in the cleaned song pop.

song_pop_cleaned <- read_csv("song_pop_cleaned.csv", col_names = TRU
E)
song_pop_cleaned

# Now, left join song_pop_cleaned to songs_artists_acoustic_combined
.

songs_artists_acoustic_pop_combined <- songs_artists_acoustic_combin
ed %>%
  left_join(song_pop_cleaned, by = "song_id")
songs_artists_acoustic_pop_combined

# Now, time to load in the cleaned song chart.

song_chart_cleaned <- read_csv("song_chart_cleaned.csv", col_names =
TRUE)
song_chart_cleaned

# Now, left join songs_artists_acoustic_pop_combined to song_chart_c
leaned.

songs_artists_acoustic_pop_chart_combined <- songs_artists_acoustic_
pop_combined %>%
  left_join(song_chart_cleaned, by = "song_id")
songs_artists_acoustic_pop_chart_combined

# Now, we want to filter the dataset, using the filter() function, s
o that only the songs from the years 2014-2018 are shown.
# We will call this new dataset modern_music_dataset.

modern_music_dataset <- songs_artists_acoustic_pop_chart_combined %>
%
  filter(year >= 2014)
modern_music_dataset

# This is our main dataset for the project, which contains 1,377 obs
ervations and 25 variables.
# Time to check that all of the variables have been correctly
# labelled by R using the glimpse() function.

glimpse(modern_music_dataset)

# R has correctly labelled all of the continuous numerical variables
as dbl, the
# explicit and is_pop variables as logical and all of the categorica
l variables
# as character. Now, to check that there are no duplicate
# song ids using the duplicated() and sum() functions.
```

```r
sum(duplicated(modern_music_dataset$song_id))

# As you can see, there are no duplicate song ids.
# Now, to check to see if there are any missing values for each vari
able in the dataset
# using the colSums(is.na()) function.

colSums(is.na(modern_music_dataset))

# Of importance for this project, there are 64 missing values in art
ist_type and
# 35 missing values in main_genre. These will be dealt with per grap
h, to minimize
# information loss.

# Now, to export modern_music_dataset to a table
# using write.table() function and the file will be called "modern_m
usic_dataset.csv".

write.table(modern_music_dataset, "modern_music_dataset.csv", sep=",
", row.names=FALSE)

# This dataset has now been saved to my files and I can now begin cr
eating the graphs.

# ************************************************************************
**********************************************
# FIGURE #1 HEAT MAP OF TOP 5 MOST POPULAR GENRES AND THEIR SONG TYP
ES AND COLOURED BY PERCENTAGE OF POPULAR SONGS
# ************************************************************************
**********************************************

# First, since our heat map is using the main_genre variable, we nee
d to remove the 35 missing values using the filter() and is.na() fun
ctions.
# We will call this new dataset modern_music_genre_cleaned.

modern_music_genre_cleaned <- modern_music_dataset %>%
  filter(!is.na(main_genre))
modern_music_genre_cleaned

# Now, I'm going to create a new dataframe called modern_music_genre
_grouped and I'm creating a new variable called main_genre_broad.

modern_music_genre_grouped <- modern_music_genre_cleaned %>%
  mutate(main_genre_broad = case_when(

    # Use str_detect to find any genre containing "pop" and record i
t as "Pop" in main_genre_broad.
```

```
    str_detect(main_genre, "pop") ~ "Pop",

    # Use str_detect to find any genre containing "hip hop, rap, or
trap" and record it as "Hip Hop" in main_genre_broad.

    str_detect(main_genre, "hip hop|rap|trap") ~ "Hip Hop",

    # Use str_detect to find any genre containing "r&b" and record i
t as "R&B" in main_genre_broad.

    str_detect(main_genre, "r&b") ~ "R&B",

    # Use str_detect to find any genre containing "soul" and record
it as "Soul" in main_genre_broad.

    str_detect(main_genre, "soul") ~ "Soul",

    # Use str_detect to find any genre containing "edm, house, elect
ro, brostep, big room" and record it as "EDM" in main_genre_broad.

    str_detect(main_genre, "edm|house|electro|brostep|big room") ~ "
EDM",

    # Use str_detect to find any genre containing "country" and reco
rd it as "Country" in main_genre_broad.

    str_detect(main_genre, "country") ~ "Country",

    # Use str_detect to find any genre containing "rock" and record
it as "Rock" in main_genre_broad.

    str_detect(main_genre, "rock") ~ "Rock",

    # Use str_detect to find any genre containing "metal" and record
it as "Metal" in main_genre_broad.

    str_detect(main_genre, "metal") ~ "Metal",

    # Use str_detect to find any genre containing "indie" and record
it as "Indie" in main_genre_broad.

    str_detect(main_genre, "indie") ~ "Indie",

    # Use str_detect to find any genre containing "folk" and record
it as "Folk" in main_genre_broad.

    str_detect(main_genre, "folk") ~ "Folk",

    # Use str_detect to find any genre containing "dance" and record
it as "Dance" in main_genre_broad.
```

```r
    str_detect(main_genre, "dance") ~ "Dance",

    # Keep everything else as its original title-cased name

    TRUE ~ stringr::str_to_title(main_genre)

    ))

# Now, to find the top 5 genres with the most popular songs and call
the dataset top_5_popular_genres.

top_5_popular_genres <- modern_music_genre_grouped %>%
  filter(is_pop == TRUE) %>% # Filter to look only at popular songs
  count(main_genre_broad, sort = TRUE) %>%  # Count how many popular
songs are in each genre
  slice_head(n = 5) # Only focus on the top 5 genres.

# View the result
print(top_5_popular_genres)

# 1. Prepare Heat Map Data
heatmap_success_data <- modern_music_genre_grouped %>%
  filter(main_genre_broad %in% top_5_popular_genres$main_genre_broad
) %>% # Only select top 5 most popular genres.
  group_by(main_genre_broad, song_type) %>% # Group by main_genre_br
oad and song_type.
  summarise(total_songs = n(), # Count total songs in group
  pop_count = sum(is_pop == TRUE), # Count popular songs in group
  success_rate = pop_count / total_songs, # Calculate percentage
  .groups = 'drop') %>% # Un-group for further steps
  tidyr::complete(main_genre_broad, song_type) %>% # Ensures all Gen
re/Song Type combinations exist (adds NAs for missing data)
  mutate(success_rate = ifelse(success_rate == 0, NA, success_rate))
# Convert success_rate of 0 to NA so it triggers the grey box and "N
o Data" label
heatmap_success_data

# 2. Create The Heat map
ggplot(heatmap_success_data, aes(x = song_type, y = reorder(main_gen
re_broad, success_rate, FUN = mean, na.rm = TRUE), fill = success_ra
te)) + # Reorder Y-axis genres by their average success rate.
  geom_tile(color = "white", linewidth = 0.8) + # Draw the heatmap t
iles with a white border and certain width.
  geom_text(aes(label = ifelse(is.na(success_rate), "No Data", scale
s::percent(success_rate, accuracy = 1)), # Add text labels inside th
e tiles. Add label "No Data" for tiles that contain no data.
  color = ifelse(is.na(success_rate) | success_rate < 0.4, "black",
"white")), fontface = "bold", size = 5, family = "sans") + # Use whi
te text on dark tiles, black on light/grey tiles. Size 5 sans-serif
```

```r
font and bold.
  scale_fill_gradient(low = "#f7fbff", high = "#0072B2", na.value =
"grey90", labels = scales::percent_format(accuracy = 1), name = "Pop
ularity Rate (%)", # Define the blue color gradient and legend scale
colours and giving the legend a title.
  guide = guide_colorbar(barheight = unit(6, "cm"), barwidth = unit(
0.5, "cm"))) +
  scale_color_identity() + # Ensure the black/white text colors are
applied. # Labelling the x-axis and y-axis and giving the graph a ti
tle and subtitle.
  labs(title = "Top 5 Most Popular Genres by Their Song Types (2014
- 2018)",
  subtitle = "Percentage of Songs Reaching 'Popular' Status Within E
ach Genre", x = "Song Type", y = "Genre") +
  theme_minimal(base_size = 18) + # Apply a clean theme and set base
font size.
  # Detailed theme adjustments below for visual polish.
  theme(text = element_text(family = "sans"),
      plot.title = element_text(face = "bold", size = 22, margin = m
argin(b = 10), hjust = 0.5),
      plot.subtitle = element_text(face = "italic", size = 15, margi
n = margin(b = 20), hjust = 0.5),
      axis.title = element_text(face = "bold", size = 16),
      axis.text = element_text(color = "black", size = 14),
      panel.grid = element_blank(),
      legend.title = element_text(face = "bold", size = 14),
      legend.text = element_text(face = "bold", size = 12), # Boldin
g numeric values
      legend.position = "right",
      legend.margin = margin(l = 10),
      plot.margin = margin(10, 20, 10, 10))

# 3. Exporting to PDF with dpi = 300 for high resolution.
ggsave("Heat_Map.pdf", width = 12, height = 7, dpi = 300)

# ****************************************************************
*****************************************
# FIGURE #2 LOLLIPOP CHART OF TOP 5 GENRES BY PERCENTAGE OF SONGS TH
AT ARE POPULAR
# ****************************************************************
*****************************************

# 1. Prepare the Lollipop Chart Data
lollipop_data <- modern_music_genre_grouped %>% # Create a new datas
et called lollipop_data based on the grouped music data.
  filter(main_genre_broad %in% top_5_popular_genres$main_genre_broad
) %>% # Keep only the genres that were identified as being in the to
p 5 most popular.
  group_by(main_genre_broad) %>% # Group the data by the broad genre
name for summary calculations.
```

```r
  summarise(total_songs = n(), pop_count = sum(is_pop == TRUE, na.rm
= TRUE), # Count the total number of songs in each genre. Sum the nu
mber of popular songs, ignoring missing values
  success_rate = pop_count / total_songs, .groups = 'drop') # Divide
popular songs by total to get the success percentage and then remove
the grouping.
lollipop_data

# 2. Create the Lollipop Chart
ggplot(lollipop_data, aes(x = reorder(main_genre_broad, success_rate
), y = success_rate)) + # Initialize the ggplot object, ordering the
Genre axis by the success rate value.
  geom_segment(aes(x = reorder(main_genre_broad, success_rate), xend
= reorder(main_genre_broad, success_rate), y = 0, yend = success_rat
e, linewidth = success_rate), color = "grey50") + # Draw the "sticks
" of the lollipops using line segments
  scale_linewidth_continuous(range = c(1, 2.5), guide = "none") + #
Set the thickness range for the sticks and remove the legend for lin
e width.
  geom_point(aes(fill = success_rate, size = success_rate), shape =
21, color = "grey40", stroke = 1) + # Draw the "heads" of the lollip
ops.
  scale_size_continuous(range = c(10, 16), guide = "none") + # Set t
he size range for the heads and remove the legend for the heads.
  geom_text(aes(label = scales::percent(success_rate, accuracy = 1))
, color = "black", fontface = "bold", # Add the percentage labels to
the right of the heads and selects their font and size.
  size = 6, family = "sans", hjust = -0.6) +
  scale_fill_gradient(low = "#bcd2ee", high = "#0072B2", labels = sc
ales::percent_format(accuracy = 1), name = "Popularity Rate (%)", #
Define the color gradient for the heads fill and label the legend.
  guide = guide_colorbar(barheight = unit(6, "cm"), barwidth = unit(
0.5, "cm"))) + # Size of the legend bar.
  scale_y_continuous(labels = scales::percent_format(accuracy = 1),
breaks = seq(0, 0.5, by = 0.1), limits = c(-0.03, 0.50)) + # Configu
re the Y-axis (which will be the horizontal axis after coord_flip).
  coord_flip() +
  labs(title = "Popularity Rate of the Top 5 Most Popular Genres (20
14 - 2018)", subtitle = "", x = "Genre", y = "Popularity Rate (%)")
+ # Title and labeling of the x-axis and y-axis.
  theme_minimal(base_size = 18) + #Theme below to match the other gr
aphs.
  theme(text = element_text(family = "sans", lineheight = 1.2),
        plot.title = element_text(face = "bold", size = 22, hjust = 0.
5, margin = margin(b = 10)),
        plot.subtitle = element_text(face = "italic", size = 15, hjust
= 0.5, margin = margin(b = 20)),
        axis.title = element_text(face = "bold", size = 16),
        axis.text = element_text(color = "black", size = 14),
```

```
      panel.grid.major.y = element_blank(),
      panel.grid.minor = element_blank(),
      legend.title = element_text(face = "bold", size = 14),
      legend.text = element_text(face = "bold", size = 12),
      plot.margin = margin(10, 20, 10, 10))

# 3. Exporting to PDF with dpi = 300 for high resolution.
ggsave("Lollipop_Chart.pdf", width = 12, height = 7, dpi = 300)

# ************************************************************************
*****************************************
# FIGURE #3 GROUPED BAR CHART OF EXPLICIT VS CLEAN SONGS BY POPULARI
TY
# ************************************************************************
*****************************************

# 1. Prepare the Grouped Bar Chart Data
grouped_bar_explicit_data <- modern_music_dataset %>% # Create a new
dataset for the grouped bar chart.
  count(explicit, is_pop) %>% # Counts the number of occurrences for
every combination of explicit and is_pop.
  group_by(explicit) %>% # Tells R to perform the next calculations
separately for explicit and clean songs.
  mutate(percentage = round((n / sum(n)) * 100, 0)) %>% # calculates
the percentage of popular and not popular songs within each explicit
ness category and rounds the percentage to the nearest whole number.
  ungroup() %>%
  mutate(explicit = reorder(explicit, percentage * (is_pop == TRUE))
) # Reorders the dataset based on the percentage of popular songs.
grouped_bar_explicit_data

# 2. Create the Grouped Bar Chart
ggplot(grouped_bar_explicit_data, aes(x = percentage, y = explicit,
fill = as.factor(is_pop))) + # Initializes the plot, mapping percent
ages to the X-axis, explicitness to the Y-axis, and using popularity
to determine the bar colors.
  geom_col(position = position_dodge(width = 0.8), width = 0.7) + #
Draws the bars. position_dodge() places the bars side-by-side rather
than stacked.
  geom_text(aes(label = paste0(percentage, "%")), position = positio
n_dodge(width = 0.8), hjust = -0.3, fontface = "bold", size = 5, fam
ily = "sans") + # Adds the text labels (e.g., "33%") to the bars. Al
so, the font type, size, and positioning.
  scale_y_discrete(labels = c("TRUE" = "Explicit", "FALSE" = "Clean"
)) + # Renames the Y-axis categories from logical TRUE/FALSE to more
readable "Explicit" and "Clean".
  scale_fill_manual(values = c("TRUE" = "#0072B2", "FALSE" = "#D55E0
0"), labels = c("TRUE" = "Popular", "FALSE" = "Not Popular")) + # Ma
nually assigns specific colors to the bars(blue and orange) and rena
mes the legend items to "Popular" and "Not Popular".
```

```r
  scale_x_continuous(breaks = seq(0, 100, by = 10), limits = c(0, 10
0), expand = expansion(mult = c(0, 0.05))) + # Sets the X-axis to ru
n from 0 to 100, with tick marks every 10 units.
  labs(title = "Clean Vs. Explicit Music by Popularity (2014 - 2018)
", subtitle = "", x = "Percentage of Songs (%)", y = "Explicitness",
fill = "") + # Sets the title, axis labels, and removes the legend t
itle (empty string).
  theme_bw(base_size = 18) + # Theme below to match the other graphs
.
  theme(text = element_text(family = "sans"),
        plot.background = element_rect(fill = "white", color = NA),
        panel.background = element_rect(fill = "white", color = NA),
        panel.border = element_blank(),
        axis.line = element_line(color = "black"),
        plot.title = element_text(face = "bold", size = 22, margin =
margin(b = 10), hjust = 0.5),
        plot.subtitle = element_text(face = "italic", size = 15, mar
gin = margin(b = 20), hjust = 0.5),
        axis.title.y = element_text(face = "bold", size = 16, margin
= margin(r = 15)),
        axis.title.x = element_text(face = "bold", size = 16, margin
= margin(t = 15)),
        axis.text = element_text(face = "plain", size = 12, color =
"black"),
        legend.position = "top",
        legend.justification = "center",
        legend.direction = "horizontal",
        legend.margin = margin(t = 0, b = 10),
        panel.grid.major = element_line(color = "grey92"),
        panel.grid.minor = element_blank())

# 3. Exporting to PDF with dpi = 300 for high resolution.
ggsave("Bar_Chart_Explicit.pdf", width = 12, height = 7, dpi = 300)

# ************************************************************
***************
# FIGURE #4 LINE CHART FOR PERCENTAGE OF SONGS THAT ARE EXPLICIT OR
CLEAN BY YEAR
# ************************************************************
***************

# 1. Prepare the Line Chart Data
line_chart_1_data <- modern_music_dataset %>% # Creating a new datas
et for the line chart.
  group_by(year, explicit) %>% # Organizes the data into buckets bas
ed on the year and whether the song is explicit or clean.
  summarise(count = n(), .groups = 'drop_last') %>% # Counts the num
ber of songs in each bucket; .groups = 'drop_last' keeps the data gr
ouped by year but drops the explicit group.
  mutate(percentage = round((count / sum(count)) * 100, 0)) %>% # Wi
```

```r
thin each year, it calculates the percentage of songs that are expli
cit vs. clean and rounds to the nearest whole number.
  ungroup()
line_chart_1_data

# 2. Creating the Line Chart
ggplot(line_chart_1_data, aes(x = year, y = percentage, color = expl
icit, linetype = explicit, shape = explicit)) + # It maps year to th
e horizontal axis, percentage to the vertical, and uses the explicit
status to determine colors, line styles, and point shapes.
  geom_line(linewidth = 1.2) + # Draws the trend lines with a slight
ly thicker width for better visibility.
  geom_point(size = 3) + # Adds markers at each yearly data point to
help the viewer pinpoint the exact percentage.
  geom_text(data = line_chart_1_data %>% group_by(explicit) %>% filt
er(percentage == max(percentage)), aes(label = paste0(percentage, "%
")), vjust = -1.0, fontface = "bold", size = 5, show.legend = FALSE,
family = "sans") + # This filters the data to find the highest perce
ntage point for both clean and explicit songs and places a bold perc
entage label above those peaks.
  geom_text(data = line_chart_1_data %>% filter(year == 2018), aes(l
abel = ifelse(explicit == TRUE, "Explicit", "Clean")), hjust = -0.15
, fontface = "bold", size = 5.5, show.legend = FALSE, family = "sans
") + # This looks only at the final year (2018) and adds the text "E
xplicit" or "Clean" next to the end of the lines.
  scale_color_manual(values = c("TRUE" = "#D55E00", "FALSE" = "#0072
B2")) + # Manually sets the colors for the lines (orange for Explici
t, blue for Clean).
  scale_linetype_manual(values = c("TRUE" = "solid", "FALSE" = "dash
ed")) + # Makes the "Clean" line dashed and the "Explicit" line soli
d to make them distinguishable even for colorblind readers.
  scale_shape_manual(values = c("TRUE" = 16, "FALSE" = 17)) + # Assi
gns a circle (16) to Explicit and a triangle (17) to Clean points.
  scale_x_continuous(breaks = c(2014, 2015, 2016, 2017, 2018), expan
d = expansion(mult = c(0.05, 0.2))) + # Forces the X-axis to show ev
ery year and adds extra space on the right (0.2) to fit the "Explici
t/Clean" text labels.
  scale_y_continuous(limits = c(0, 115), breaks = seq(0, 100, by = 1
0), expand = c(0, 0)) + # Sets the Y-axis from 0 to 115% (giving hea
droom for labels) and ensures the axis starts exactly at zero.
  labs(title = "Trends in Explicit Vs. Clean Music (2014 - 2018)", s
ubtitle = "Peak Percentages Labeled for Quick Reference", x = "Year"
, y = "Percentage of Songs (%)") + # Provides the main title, italic
ized subtitle, and clear axis titles.
  theme_bw(base_size = 18) + # Theme below to match the other graphs
.
  theme(text = element_text(family = "sans"),
        plot.title = element_text(face = "bold", size = 22, margin =
margin(b = 10), hjust = 0.5),
```

```r
        plot.subtitle = element_text(face = "italic", size = 15, mar
gin = margin(b = 20), hjust = 0.5),
        axis.title.y = element_text(face = "bold", size = 16, margin
= margin(r = 15)),
        axis.title.x = element_text(face = "bold", size = 16, margin
= margin(t = 15)),
        axis.text = element_text(face = "plain", size = 12, color =
"black"),
        legend.position = "none",
        panel.grid.major = element_line(color = "grey92"),
        panel.grid.minor = element_blank(),
        panel.border = element_blank(),
        axis.line = element_line(color = "black"))

# 3. Exporting to PDF with dpi = 300 for high resolution.
ggsave("Line_Chart_Explicit.pdf", width = 12, height = 7, dpi = 300)

# ****************************************************************
*********************************
# FIGURE #5 GROUPED BAR CHART FOR ARTIST TYPE BY POPULARITY
# ****************************************************************
*********************************

# 1. Clean Data
modern_music_artist_type_cleaned <- modern_music_dataset %>%
  filter(!is.na(artist_type)) # Removing all the missing values for
artist type.

# 2. Prepare the Grouped Bar Chart Data
grouped_bar_artist_data <- modern_music_artist_type_cleaned %>%
  count(artist_type, is_pop) %>% # Counts how many songs belong to e
ach artist type and whether they are popular or not.
  group_by(artist_type) %>% # Groups the data so the next percentage
calculation happens within each artist category.
  mutate(percentage = round((n / sum(n)) * 100, 0), artist_type = st
ringr::str_to_title(artist_type), artist_type = ifelse(artist_type =
= "Dj", "DJ", artist_type)) %>% # Calculates the percentage of popul
ar vs. not popular songs. Converts text to "Title Case" (e.g., "solo
" becomes "Solo"). Specifically fixes the capitalization of "DJ" so
it's not "Dj".
  ungroup() %>%
  mutate(artist_type = reorder(artist_type, percentage * (is_pop ==
TRUE))) # Sorts the dataset, so the artist type with the highest per
centage of "Popular" songs appears at the top.
grouped_bar_artist_data

# 3. Create the Grouped Bar Chart
ggplot(grouped_bar_artist_data, aes(x = percentage, y = artist_type,
fill = as.factor(is_pop))) + # Maps percentages to the X-axis, artis
t type to the Y-axis, and popularity to the color of the bars.
```

```r
  geom_col(position = position_dodge(width = 0.8), width = 0.7) + # Creates the bars (side-by-side) instead of stacked ones.
  geom_text(aes(label = paste0(percentage, "%")), position = position_dodge(width = 0.8), hjust = -0.3, fontface = "bold", size = 5, family = "sans") + # Ensures the text labels align with the center of each individual bar.
  scale_fill_manual(values = c("TRUE" = "#0072B2", "FALSE" = "#D55E00"), labels = c("TRUE" = "Popular", "FALSE" = "Not Popular")) + # Uses a colorblind-friendly palette (Blue/Orange) and sets the labels for the legend.
  scale_x_continuous(breaks = seq(0, 100, by = 10), limits = c(0, 110), expand = expansion(mult = c(0, 0.05))) + # Sets the scale from 0 to 110 (the extra 10 provides space for the text labels so they aren't cut off).
  labs(title = "Artist Types By Popularity (2014 - 2018)", subtitle = "", x = "Percentage of Songs (%)", y = "Artist Type", fill = "") + # Defines the title, subtitle, and axis names.
  theme_bw(base_size = 18) + # Theme below to match the other graphs.
  theme(text = element_text(family = "sans"),
        plot.background = element_rect(fill = "white", color = NA),
        panel.background = element_rect(fill = "white", color = NA),
        panel.border = element_blank(),
        axis.line = element_line(color = "black"),
        plot.title = element_text(face = "bold", size = 22, margin = margin(b = 10), hjust = 0.5),
        plot.subtitle = element_text(face = "italic", size = 15, margin = margin(b = 20), hjust = 0.5),
        axis.title.y = element_text(face = "bold", size = 16, margin = margin(r = 15)),
        axis.title.x = element_text(face = "bold", size = 16, margin = margin(t = 15)),
        axis.text = element_text(face = "plain", size = 12, color = "black"),
        legend.position = "top",
        legend.justification = "center",
        legend.direction = "horizontal",
        legend.margin = margin(t = 0, b = 10),
        panel.grid.major = element_line(color = "grey92"),
        panel.grid.minor = element_blank())

# 4. Exporting to PDF with dpi = 300 for high resolution.
ggsave("Bar_Chart_Artist_Type.pdf", width = 12, height = 7, dpi = 300)
```